

Higher genus theory and unramified quadratic extensions of biquadratic fields

Arithmetic Statistics, Paderborn

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joint work with Siu Hang Man and Pavlo Yatsyna

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27 August 2026

Unit signature ranks

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- The collection of all signatures of units $\alpha \in \mathcal{O}_K^\times$ can be viewed as a subspace of $\mathbb{F}_2^n$.
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Question: How is $\text{sgnrk}(K)$ distributed amongst some family of totally real fields K ?

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- If D is not divisible by any prime $p \equiv 3 \pmod{4}$, then $\text{sgnrk}(K) \in \{1, 2\}$. But we still find many examples in both cases:

$$\text{sgnrk}(K) = 1 : K = \mathbb{Q}(\sqrt{34}), \mathbb{Q}(\sqrt{146}), \mathbb{Q}(\sqrt{178}), \mathbb{Q}(\sqrt{194}), \mathbb{Q}(\sqrt{205}), \dots$$

$$\text{sgnrk}(K) = 2 : K = \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{10}), \mathbb{Q}(\sqrt{13}), \mathbb{Q}(\sqrt{17}), \mathbb{Q}(\sqrt{26}), \dots$$

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Which case occurs more often?

Real quadratic fields: a brief history

Define the sets:

$$\mathcal{D} := \{D > 0 \text{ squarefree} : p \mid D \Rightarrow p \not\equiv 3 \pmod{4}\},$$

$$\mathcal{D}^- := \{D > 0 \text{ squarefree} : \text{sgnrk } \mathbb{Q}(\sqrt{D}) = 2\}.$$

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- (Nagell 1932) Conjectured that the limit

$$\lim_{X \rightarrow \infty} \frac{\#\mathcal{D}^-(X)}{\#\mathcal{D}(X)}$$

exists and lies in the open interval $(0, 1)$.

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- (Fouvry–Klüners 2010) Improved their earlier lower bound to

$$\liminf_{X \rightarrow \infty} \frac{\#\mathcal{D}^-(X)}{\#\mathcal{D}(X)} \geq \frac{5}{4}\alpha \approx 0.52475\dots$$

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- (Chan–Koymans–Milovic–Pagano 2019) Used the results of Smith to further improve the lower bound

$$\liminf_{X \rightarrow \infty} \frac{\#\mathcal{D}^-(X)}{\#\mathcal{D}(X)} \geq \alpha \cdot \sum_{n=0}^{\infty} 2^{-n(n+3)/2} \approx 0.538 \dots$$

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- (Koymans–Pagano 2022) Proved Stevenhagen's conjecture holds, i.e. the limit exists and is equal to

$$\lim_{X \rightarrow \infty} \frac{\#\mathcal{D}^-(X)}{\#\mathcal{D}(X)} = 1 - \alpha \approx 0.5805 \dots \quad \text{☺}$$

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The sequence $\{\mathrm{rk}_{2^m}(G)\}_{m \geq 1}$ uniquely determines the 2-primary part of G .

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- For any K totally real of degree n :

$$\begin{aligned} \mathrm{sgnrk}(K) = n &\iff \mathrm{Cl}(K) \cong \mathrm{Cl}^+(K) \\ &\iff \mathrm{rk}_{2^m} \mathrm{Cl}(K) = \mathrm{rk}_{2^m} \mathrm{Cl}^+(K) \text{ for all } m \geq 1 \end{aligned}$$

since the odd parts of $\mathrm{Cl}(K)$ and $\mathrm{Cl}^+(K)$ agree.

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- Thus, if for example either $\mathrm{rk}_4 \mathrm{Cl}^+(K) = 0$, or $\mathrm{rk}_8 \mathrm{Cl}^+(K) = 0$ and $\mathrm{rk}_4 \mathrm{Cl}^+(K) = \mathrm{rk}_4 \mathrm{Cl}(K)$, then $\mathrm{sgnrk}(K) = 2$.

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To therefore obtain bounds for the ratio $\#\mathcal{D}^-(X)/\#\mathcal{D}(X)$, we must count fields K with prescribed $\mathrm{rk}_4 \mathrm{Cl}^+(K)$, $\mathrm{rk}_8 \mathrm{Cl}^+(K)$, etc.

Rédei matrices

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1. For $i \neq j$, we define $R(D)_{ij} := r_{ij}$ where $\left(\frac{p_i}{p_j}\right) = (-1)^{r_{ij}}$.
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Theorem (Rédei 1934)

Let $K = \mathbb{Q}(\sqrt{D})$ with $D \in \mathcal{D}$. Then

$$\mathrm{rk}_4 \mathrm{Cl}^+(K) = \omega(D) - 1 - \mathrm{rank}_{\mathbb{F}_2} R(D) \quad (= \mathrm{corank}_{\mathbb{F}_2} R(D) - 1)$$

Rédei matrices: an example

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so all off-diagonal entries equal 1, and all diagonal entries equal 0:

$$R(2405) = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

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We compute $\text{rank } R(D) = 2$ and $\text{corank } R(D) = 1$. Hence

$$\text{rk}_2 \text{Cl}^+ = 2, \quad \text{rk}_4 \text{Cl}^+ = 0, \quad \text{Cl}^+(K) \cong C_2 \times C_2,$$

and in particular $x^2 - 2405y^2 = -1$ is soluble.

Counting Rédei Matrices

Let $\mathcal{R}_t$ be the set of all Rédei matrices of size $t \times t$. We have a rank-preserving bijection from $\mathcal{R}_t$ to the set of symmetric $(t-1) \times (t-1)$ matrices over $\mathbb{F}_2$

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Theorem (MacWilliams 1969)

Let $N(t, r)$ be the number of symmetric $t \times t$ matrices over $\mathbb{F}_2$ of rank r . Then:

$$N(t, 2s) = \prod_{i=1}^s \frac{2^{2i}}{2^{2i} - 1} \cdot \prod_{i=0}^{2s-1} (2^{t-i} - 1), \quad N(t, 2s+1) = \prod_{i=1}^s \frac{2^{2i}}{2^{2i} - 1} \cdot \prod_{i=0}^{2s} (2^{t-i} - 1).$$

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Corollary

Fix some non-negative integer $m \geq 0$. Then

$$\lim_{t \rightarrow \infty} \frac{\#\{M \in \mathcal{R}_t \mid \text{corank}_{\mathbb{F}_2} M = m+1\}}{\#\mathcal{R}_t} = \alpha \prod_{i=1}^m (2^i - 1)^{-1}$$

Biquadratic fields

Let $K = \mathbb{Q}(\sqrt{d_1}, \sqrt{d_2})$ be a real biquadratic field, with quadratic subfields $K_i = \mathbb{Q}(\sqrt{d_i})$, where $d_3 = d_1 d_2 / \gcd(d_1, d_2)^2$. Here $n = 4$.

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- By a theorem of Edgar–Mollin–Peterson (1986), we have

$$\begin{aligned} \text{sgnrk}(K) = 4 &\implies \text{sgnrk}(K_i) = 2 && \text{for all } i = 1, 2, 3 \\ &\implies x^2 - D_i y^2 = -1 \text{ solvable} && \text{for all } i = 1, 2, 3. \end{aligned}$$

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- (Kubota 1956) If $\text{sgnrk}(K_i) = 2$ for each $i = 1, 2, 3$, then $\text{sgnrk}(K) \in \{3, 4\}$.

We again find many examples in both cases:

$$\text{sgnrk}(K) = 3 : K = \mathbb{Q}(\sqrt{2}, \sqrt{85}), \mathbb{Q}(\sqrt{10}, \sqrt{17}), \mathbb{Q}(\sqrt{13}, \sqrt{85}), \mathbb{Q}(\sqrt{17}, \sqrt{65}), \dots$$

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Rédei matrices for biquadratic fields

Write $\Delta_K = D^2$ with $D \in \mathcal{D}$, and let $p_1, \dots, p_t$ be the primes dividing D . They partition into three blocks

$$\gamma_1 = \gcd(d_1, d_3), \quad \gamma_2 = \gcd(d_2, d_3), \quad \gamma_3 = \gcd(d_1, d_2),$$

of sizes t_1, t_2, t_3 (so $d_1 = \gamma_1\gamma_3$, $d_2 = \gamma_2\gamma_3$, $d_3 = \gamma_1\gamma_2$).

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of sizes t_1, t_2, t_3 (so $d_1 = \gamma_1\gamma_3$, $d_2 = \gamma_2\gamma_3$, $d_3 = \gamma_1\gamma_2$). Let $\underline{R}(D)$ be the Rédei *prematrix* of D consisting of Jacobi symbols (p_i/p_j) . We break $\underline{R}(D)$ into blocks,

$$\underline{R}(D) = \begin{pmatrix} \underline{A}_{11} & A_{12} & A_{13} \\ A_{21} & \underline{A}_{22} & A_{23} \\ A_{31} & A_{32} & \underline{A}_{33} \end{pmatrix},$$

where the Rédei prematrices of D_i are given by

$$\underline{R}(D_1) = \begin{pmatrix} \underline{A}_{11} & A_{13} \\ A_{31} & \underline{A}_{33} \end{pmatrix}, \quad \underline{R}(D_2) = \begin{pmatrix} \underline{A}_{22} & A_{23} \\ A_{32} & \underline{A}_{33} \end{pmatrix}, \quad \underline{R}(D_3) = \begin{pmatrix} \underline{A}_{11} & A_{12} \\ A_{21} & \underline{A}_{22} \end{pmatrix}.$$

The corresponding Rédei matrices $R(D_i)$ are obtained by filling in the diagonals of $\underline{R}(D_i)$.

Rédei matrices for biquadratic fields

Theorem (Man–V–Yatsyna)

Let $m_1, m_2, m_3 \in \mathbb{N}_0$ be fixed integers. Then, as $t \rightarrow \infty$ with $t_1 \geq t_2 \geq t_3 \geq \frac{t}{10}$, we have

$$\lim_{t \rightarrow \infty} \frac{\#\{A \in \mathcal{R}_t \mid \text{corank}_{\mathbb{F}_2} A_{[i]} = m_i + 1, i = 1, 2, 3\}}{\#\mathcal{R}_t} = \alpha^3 \prod_{i=1}^3 \prod_{j=1}^{m_i} (2^j - 1)^{-1},$$

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Construction of $A_{[i]}$:

1. Break down the prematrix $\underline{A}$ associated to A into blocks of size t_1, t_2, t_3 respectively

$$\underline{A} = \begin{pmatrix} \underline{A}_{11} & A_{12} & A_{13} \\ A_{21} & \underline{A}_{22} & A_{23} \\ A_{31} & A_{32} & \underline{A}_{33} \end{pmatrix},$$

2. Take $A_{[i]}$ to be Rédei matrices:

$$A_{[1]} = R \begin{pmatrix} \underline{A}_{11} & A_{13} \\ A_{31} & \underline{A}_{33} \end{pmatrix}, \quad A_{[2]} = R \begin{pmatrix} \underline{A}_{22} & A_{23} \\ A_{32} & \underline{A}_{33} \end{pmatrix},$$
$$A_{[3]} = R \begin{pmatrix} \underline{A}_{11} & A_{12} \\ A_{21} & \underline{A}_{22} \end{pmatrix}.$$

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$$m - 1 \text{ w.p. } 1 - 2^{-m}, \quad m \text{ w.p. } 2^{-m-1}, \quad m + 1 \text{ w.p. } 2^{-m-1}.$$

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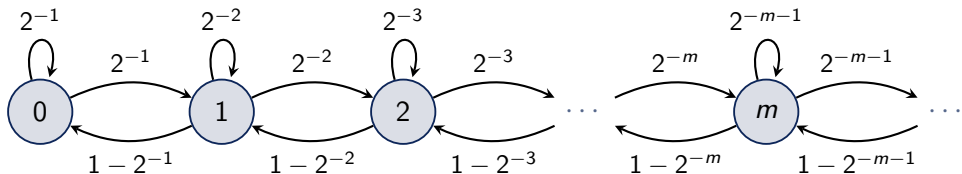
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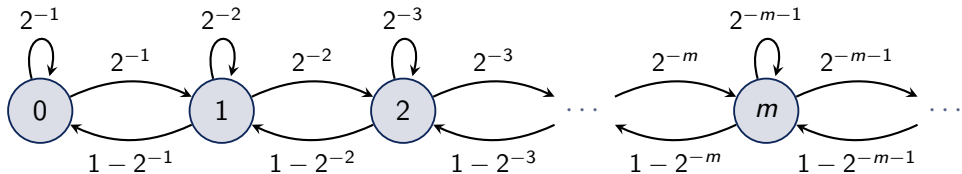
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4. **Independence:** Do the above to track the coranks of each of $A_{[1]}$, $A_{[2]}$, and $A_{[3]}$, and show these distributions are independent.

Proof strategy



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The probability of landing on corank m approaches $\alpha \prod_{j=1}^m (2^j - 1)$.

Back to fields

Let $\mathcal{K}(X) := \{K \text{ real biquadratic} \mid \Delta_K = D^2, D \in \mathcal{D}(X)\}$

Theorem (Man-V-Yatsyna)

$$\liminf_{X \rightarrow \infty} \frac{\#\{K \in \mathcal{K}(X) \mid N_{K_i/\mathbb{Q}}(\varepsilon_i) = -1, \forall 1 \leq i \leq 3\}}{\#\mathcal{K}(X)} \geq \alpha^3 \approx 0.0738 \dots$$

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- This is only a lower bound, as we have contented ourselves with the case $\text{rk}_4 \text{Cl}^+(K) = 0$ (though the method extends to positive rk_4 , as in Fouvry-Klüners and CKMP)

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 - (i) almost all $D \in \mathcal{D}_t(X)$ is contained in some W_i ; the exceptions have relative size $\ll (\log \log X)^{-c}$ for some small $c > 0$.
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3. For fixed $D \in \mathcal{D}_t(X)$, almost all biquadratic K with $\Delta_K = D^2$ satisfy $t_1, t_2, t_3 \geq \frac{t}{10}$. The exceptions have relative size $\ll e^{-c \log \log X}$ for some small $c > 0$.

Kuroda's formula

Theorem (Kuroda's formula)

Let K be real biquadratic with quadratic subfields K_1, K_2, K_3 . Then

$$h(K) = \frac{Q}{4} \cdot h(K_1)h(K_2)h(K_3), \quad \text{where } Q := [\mathcal{O}_K^\times : \mathcal{O}_{K_1}^\times \mathcal{O}_{K_2}^\times \mathcal{O}_{K_3}^\times].$$

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Suppose $h^+(K_i) = h(K_i)$ for $i = 1, 2, 3$. Then

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$$\text{sgnrk}(K) = 4 \iff h^+(K) = h(K) \iff H_2^+(K) \text{ totally real.}$$

From class numbers to fields

To obtain bounds on the number of fields with $\text{sgnrk}(K) = 4$, we make the following simplifying assumption:

$$\text{rk}_4 \text{Cl}^+(K_1) = \text{rk}_4 \text{Cl}^+(K_2) = \text{rk}_4 \text{Cl}^+(K_3) = 0. \quad (*)$$

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To determine if $\text{sgnrk}(K) = 4$, we need to know if $h_2(K) = h_2^+(K)$, i.e. if the two narrow class field $H_2^+(K)$ is *totally real*. We do this by classifying quadratic extensions of K which are unramified at all finite places (UAFP) and determining which are real.

Quadratic UAFP extensions

Recall $\text{rk}_2 \text{Cl}^+(K) = \#\{\text{independent UAFP quadratic extensions of } K\}$.

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But there are more!

Example

For $K = \mathbb{Q}(\sqrt{130}, \sqrt{493})$, the extension $K(\sqrt{\beta})/K$ is UAFP, where

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Can we count how many there are, and can we explicitly classify them ?

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To do this, we use the **higher genus theory** of Koymans and Pagano.

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The theory gives us a way to handle $\text{Cl}^+(K)[2]$. Unramified $\mathbb{F}_2$ -extensions of K are built from 2-cocycles assembled out of data on the multiquadratic subfields of index 2 (the “corners” of K), and whether such a cocycle is trivial in $H^2(G_{\mathbb{Q}}, \mathbb{F}_2)$ is a condition on local Hilbert symbols.

Classifying quadratic UAFP extensions

Let $\gamma_1, \gamma_2, \gamma_3 \in \mathcal{D}$ be pairwise coprime, and let $d_1 = \gamma_1\gamma_3$, $d_2 = \gamma_2\gamma_3$, $d_3 = \gamma_1\gamma_2$. Let $K = \mathbb{Q}(\sqrt{d_1}, \sqrt{d_2})$. From higher genus theory, the unramified non-genus $\mathbb{F}_2$ -extensions of K are characterised by 2-cocycles $\theta : G_{\mathbb{Q}} \rightarrow \mathbb{F}_2$ of the form

$$\theta = \chi_{d_1} \cup \chi_{b_2 b_3} + \chi_{d_2} \cup \chi_{b_1 b_3}, \quad b_i \mid \gamma_i,$$

such that θ is trivial in $H^2(G_{\mathbb{Q}}, \mathbb{F}_2)$, i.e. θ is a *coboundary*.

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- The Fouvry–Klüners extensions are exactly the *pure* cocycles, where only one of b_1, b_2, b_3 is nontrivial (hence why these miss some UAFP extensions).

Counting UAFP extensions

Define the auxiliary Rédei matrix R' as the symmetric $\mathbb{F}_2$ -matrix with entries:

$$w_{p,q} = \begin{cases} \ell, & q \mid \gamma_j, j \neq i, \left(\frac{q}{p}\right) = (-1)^\ell, \\ 0, & q \mid \gamma_i, q \neq p, \\ \tau, & q = p, \left(\frac{\prod_{j \neq i} \gamma_j}{p}\right) = (-1)^\tau. \end{cases}$$

i.e. R' is the Rédei matrix of D with the intra-block off-diagonal entries deleted and diagonals recomputed such that rows sum to 0.

Theorem

The number of independent non-genus UAFP C_2 -extensions of K is $\text{corank } R' - 1$, and

$$\text{rk}_2 \text{Cl}^+(K) = \omega(K) + \text{corank}(R') - 3.$$

Example

Let $d_1 = p_1 p_2$, $d_2 = p_3 p_4$. Under our assumption $\text{rk}_4 \text{Cl}^+(K_i) = 0$, up to permuting the primes, the Rédei prematrix $A_K = \left(\left(\frac{p_i}{p_j} \right) \right)_{ij}$ is either

$$\begin{pmatrix} * & 1 & 1 & 0 \\ 1 & * & 0 & 0 \\ 1 & 0 & * & 1 \\ 0 & 0 & 1 & * \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} * & 1 & 1 & 1 \\ 1 & * & 0 & 0 \\ 1 & 0 & * & 1 \\ 1 & 0 & 1 & * \end{pmatrix}.$$

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Deleting the intra-block entries ($p_1 p_2$ and $p_3 p_4$) and refilling the diagonals gives

$$R' = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \text{ corank} = 3, \quad \text{resp.} \quad R' = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \text{ corank} = 2.$$

With $\omega(K) = 4$ the theorem gives $\text{Cl}_2^+(K) \cong C_2^4$, resp. $C_4 \times C_2^2$.

But this is not yet enough

Two fields with the same Legendre data

Take $\gamma_1 = 5$, $\gamma_2 = 13$ and $\gamma_3 = 37$ or 97 . In both cases all three Legendre symbols equal -1 , so A_K is the same, and indeed $\text{Cl}_2^+(K) \cong C_4$ in both cases. But:

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- Instead of only Legendre symbols, we must use **Rédei symbols**.

Detecting real UAFP extensions

Let $\theta = \chi_{d_1} \cup \chi_{b_2 b_3} + \chi_{d_2} \cup \chi_{b_1 b_3}$ be a 2-cocycle as above, and let L_θ/K be the corresponding UAFP extension. From higher genus theory, we have that $\theta = d\psi$ is the coboundary of a 1-cochain, whose field of definition $L(\psi)$ is an unramified $\mathbb{F}_2$ -extension of the multiquadratic field $M_\theta = \mathbb{Q}(\sqrt{d_1}, \sqrt{d_2}, \sqrt{b_1 b_3}, \sqrt{b_2 b_3})$.

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- For Fourvy–Klüner extensions, coming from a decomposition of second type $D = D_1 D_2$, realness is determined by the single Rédei symbol $[D_1, D_2, -1]$.
- For non-genus non-FK extensions, this unfortunately isn't determined by a single Rédei symbol. We must in general take a sum of Rédei symbols.

The criterion

Theorem (Man–V–Yatsyna)

Let R be the Rédei matrix of $\mathbb{Q}(\sqrt{\gamma_1\gamma_2\gamma_3})$ and R' the auxiliary matrix above, and let $t_k = \omega(\gamma_k)$. Let $\emptyset \subsetneq B \subsetneq [t]$ be a subset such that $\sum_{i \in B} e_i \in \ker R'$, and let $b_i = \gcd(\prod_{j \in B} p_j, \gamma_i)$.

Then $\theta = \chi_{d_1} \cup \chi_{b_2 b_3} + \chi_{d_2} \cup \chi_{b_1 b_3}$ is trivial in $H^2(G_{\mathbb{Q}}, \mathbb{F}_2)$, and there exist subsets $C_i, D_i \subseteq [t]$ depending only on t_1, t_2, t_3, R and B (and not on the primes) such that, writing $u_i = \prod_{j \in C_i} p_j$ and $v_i = \prod_{j \in D_i} p_j$, the field L_θ is totally real if and only if

$$\sum_i [u_i, v_i, -1] = 0 \in \mathbb{F}_2.$$

What we expect to prove

Theorem? (Man–V–Yatsyna 2026+)

A positive density of real biquadratic fields, ordered by discriminant, whose quadratic subfields all have unit signature rank 2, have unit signature rank 3.

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Conjecture

100% of real biquadratic fields, ordered by discriminant, whose quadratic subfields all have unit signature rank 2, have unit signature rank 3.

What remains

- Can we model the distribution of the corank(R') for the auxiliary Rédei matrix R' , i.e. the distribution of $\text{rk}_2 \text{Cl}^+(K)$? (this is almost done!)
- Can we remove the simplifying hypothesis that $\text{rk}_4 \text{Cl}^+(K_i) = 0$ for every $i = 1, 2, 3$, i.e. control $\text{rk}_8 \text{Cl}^+(K)$ and characterise UAFP extensions in this more general setting.
- Can we count and classify the cyclic quartic UAFP extensions? (this will be needed to attack the strong conjecture).

Thank you!